

Machine-Learning-Enhanced Analysis of the Ising Model on Hyperbolic Lattices

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(Dated: December 2024)

Owing to their unique geometry, visualization of hyperbolic lattices on Euclidean surfaces necessitates exotic representations that visually distort the regular patterning of the system. The semantics behind the visualization of these systems are typically uninteresting from the viewpoint of computation, but the application of machine learning models onto these systems gives new meaning to this problem. Namely, while Euclidean systems benefit from powerful machine learning techniques like convolutional neural networks, new approaches must be formulated in order to implement high-performance machine learning models onto hyperbolic lattice problems. Here, the Ising model on a hyperbolic lattice is studied. While completely clean results displaying the paramagnetic to ferromagnetic transition are readily interpreted by simple machine learning models, it will be shown that the formation of domain walls can quickly destroy the efficacy of these models. To recover the performance of the machine learning model, graph convolutional neural networks are shown to dramatically improve performance, allowing machine learning models to learn both global and local properties on hyperbolic lattices, despite their geometry.

I. BACKGROUND THEORY

A. The Ising Model

The Ising model is one of the simplest models that exhibits a phase transition.

$$H = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j \quad (1)$$

The Hamiltonian governing the model is given by Eq.(1), where J is the spin-spin coupling coefficient, σ_i is the Ising spin ($\equiv \pm 1$) on the i th site, and the sum runs over only nearest-neighbor sites. Physically, the Ising model is significant due to the fact that it embodies the physics behind the paramagnetic (PM) to ferromagnet (FM)/anti-ferromagnetic (AFM) transition. The nature of the ordered phase (FM or AFM) is determined by the sign of the spin-coupling coefficient, where $J > 1$ and $J < 1$ correspond to the FM and AFM ordered phases, respectively.

$$m = \frac{1}{N} \sum_i \sigma_i \quad (2)$$

To quantize the FM order, magnetization (m), given by Eq.(2), is used as the order parameter, where N is the total number of sites in the system. When magnetization is zero ($m = 0$), the system is said to be in the PM phase. Meanwhile, when magnetization is equal to plus or minus one ($m = \pm 1$), the system is said to be in the FM phase. In practice, m^2 may also be used as an order parameter (like it is later in this work) since then a FM phase is described by the order parameter equal to +1 instead of ± 1 .

These definitions of magnetization/magnetization-squared as the order parameter for FM order are not sufficient when considering a FM phase with domain walls, because in these systems a system-wide average

of the spins can give any number between zero and one. Thus, special consideration must be taken when considering such a case; a blind calculation of magnetization or magnetization-squared can be misleading.

In this study, only the FM ($J > 1$) case is considered, however the general principles and results are easily generalizable to the AFM case as well.

B. Hyperbolic Lattices

Hyperbolic lattices are a class of materials, defined as regular tilings of a constant negative curvature surface. In this way, hyperbolic lattices are analogous to a system like graphene, which is just a tiling of a Euclidean surface via hexagons. However, since our world is Euclidean (at

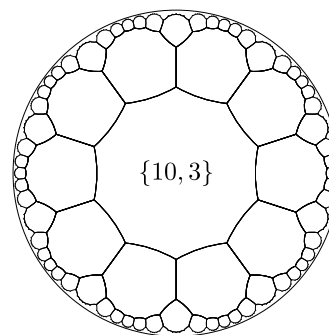


FIG. 1. Poincaré disk representation of a $\{10,3\}$ hyperbolic lattice. The notation used here is the Schläfli symbol $\{p,q\}$, where p is the number of sides of the fundamental polygon and q is the coordination number. This specific hyperbolic lattice (with three generations) is the system studied throughout this paper.

least at our length scales), special steps need to be taken when studying hyperbolic lattices. Perhaps most striking of these details is how hyperbolic lattices are visualized.

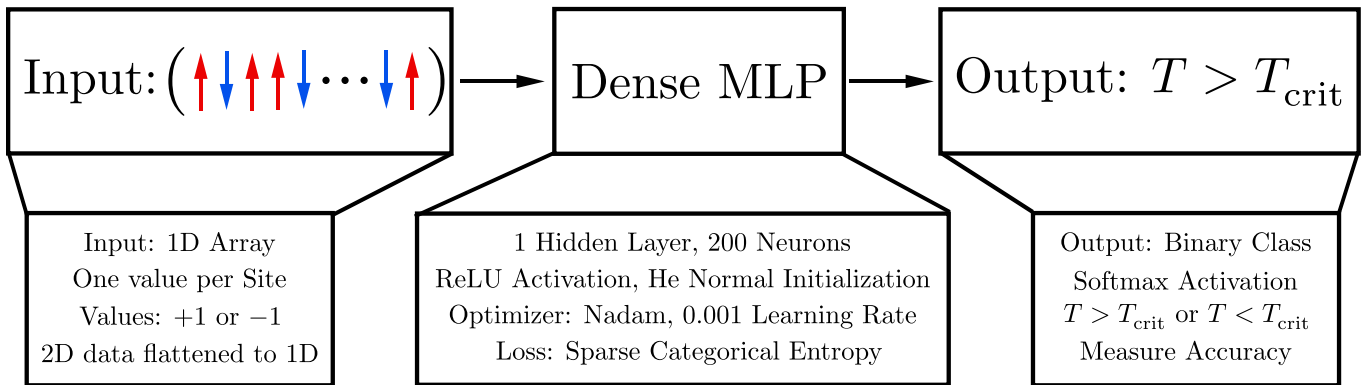


FIG. 2. Diagram depicting the specific architecture of the FCNN used in this work

Fig. 1 represents one method of visualizing a hyperbolic lattice on a Euclidean surface (such as the paper this work is printed on), called the Poincaré disk representation. Despite its appearance, the lattice illustrated in Fig. 1 is indeed regular in its native hyperbolic space. Special attention should be paid to the edge of this construction. It can be seen that the edge of the hyperbolic disk features a large number of sites packed closely together, which is a visual artifact of using the Poincaré disk representation. These core features of how hyperbolic lattices are illustrated on paper will play a crucial role in later discussion on how machine learning models can be applied to this system. Another detail to consider when studying hyperbolic lattices is their boundary. While Euclidean lattices benefit from relatively simple periodic boundary conditions (PBC), hyperbolic lattices have highly-non-trivial PBCs[1]. As a consequence of this, hyperbolic lattices with open boundary conditions (OBC) are studied here.

II. METHODOLOGY

A. Classical Monte Carlo

Data for the Ising model on a $\{10, 3\}$ hyperbolic lattice is computed using the well-known Metropolis algorithm. To speed-up convergence, expectation-informed initial conditions can be imposed. Since FM order is expected from the model, the initial starting configuration for the Monte Carlo simulation can be chosen so that most of the spins in the configuration are $\sigma_i = +1$ or $\sigma_i = -1$. Apart from speeding-up convergence, these initial conditions also promote FM states without domain walls. Later, when domain walls are specifically studied, the initial condition of the Monte Carlo computation is set to be completely random. Notably, no specific order parameter is derived here to quantify if a system with domain walls is indeed in a FM state. Instead, once a critical temperature (T_{crit}) value is identified, temperatures well below it ($T \ll T_{\text{crit}}$) are used for the Monte

Carlo simulation that produces the domain wall data. Due to this temperature choice, all sampled systems with domain walls should be in a FM state. The highest temperature simulated to have domain walls was $T = 0.1$, which is almost a whole order of magnitude lower than the critical temperature (discussed below) of $T_{\text{crit}} = 0.9$. Another noteworthy concern is the lack of a rigorous, explicit check to see if the data has domain walls for every sample. While this could introduce some error in the form of some data, thought to be domain wall FM states, actually being just completely clean (all spin +1 or all spin -1) FM states, the results presented below would be extremely curious if this were the case. Namely, it will be shown that models that otherwise classify clean FM states almost perfectly, completely fail when classifying data that are postulated to have domain walls. Combine this failure of the machine learning model with the underlying assumption that all data is indeed in a FM state due to the very low temperature the system is simulated at, it would seem that the underlying assumption that these system do have domain walls is justified. Additionally, while in principle magnetization can be used to quantify whether each Monte Carlo sampled data point (with no domain walls) is labeled correctly (FM or PM), such a check is not explicitly done here. This can also introduce some errors from using the Monte Carlo data to train and test a machine learning model. However, since a large number of Monte Carlo steps are simulated, the number of mislabeled points (if any) should make up a small proportion of the actual data used. With expectation-informed initial conditions, Monte Carlo simulations with 1×10^7 steps are utilized, with data sampled every 2×10^3 steps. With all-random initial conditions, 5×10^6 steps are computed, with data sampled every 1×10^3 steps.

B. Fully-Connected Neural Network

The first machine learning model method considered is the fully-connected neural network (FCNN), with one hidden layer. Such a simple machine learning model is

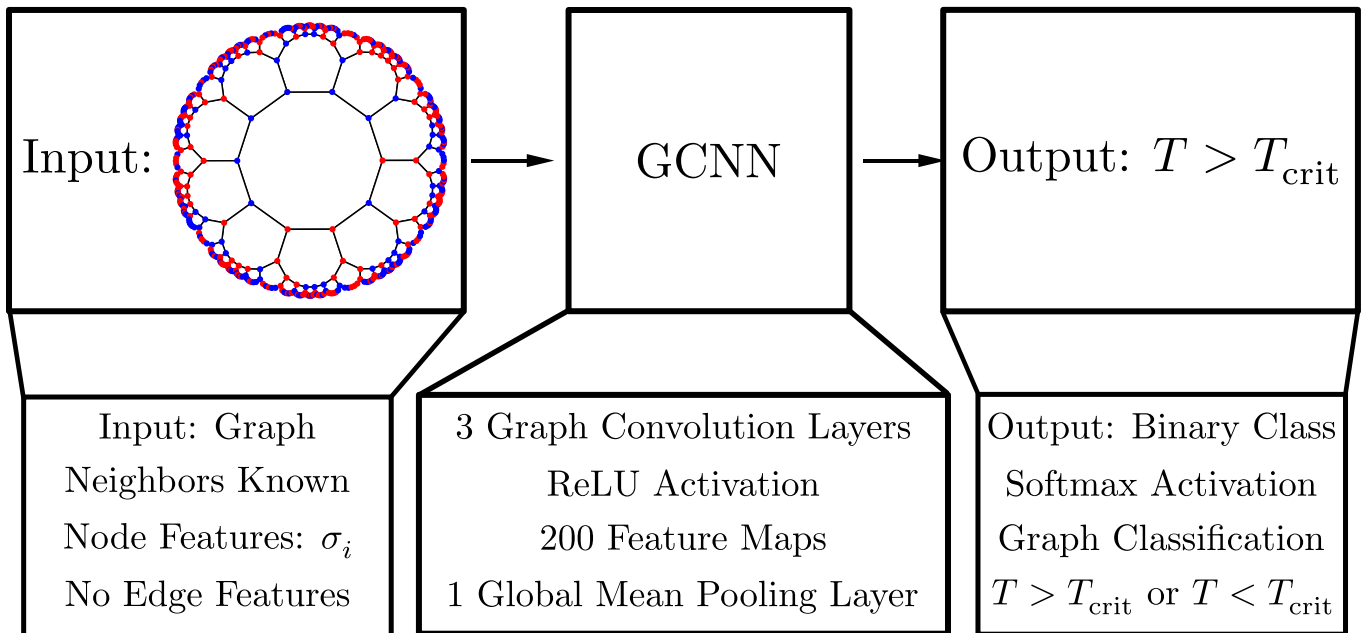


FIG. 3. Diagram depicting the specific architecture of the GCNN used in this work. The implementation of this model is through the PyG package[2]. The specific architecture of this network is heavily based on the example GCNN described in the PyG documentation[5]

justified by the simplicity of the Ising model. The input of the FCNN is a 1D-array containing the spins (σ_i) on each site of the lattice. The output is a binary classification of whether the inputted spin configuration was taken at a temperature below or above the critical temperature (T_{crit}). The specific features of the model are shown in Fig. 2, and reported in the code, made available in the supplementary.

C. Graph Convolutional Neural Network

Experience with traditional machine learning models have shown that many different architectures can be utilized to improve on the basic FCNN. Convolutional neural networks are some of the most notable of these architectures, achieving extremely high accuracy on datasets such as the MNIST dataset. Ideally, convolutional neural networks would be utilized here, like they have been in other similar studies of spin models on other lattice systems[3, 4]. However, the non-trivial geometry introduced by the hyperbolic lattice makes this impossible. While traditional convolutional neural networks cannot be used here, graph convolutional neural networks (GCNN) can be alternatively implemented. The architecture of the GCNN considered here is shown in Fig. 3. Summarily, it is composed of three graph convolutional layers, one global pooling layer, and a binary classification output via a softmax activation function. The input to the GCNN is a graph with the connectivity of the hyperbolic lattice. There is a single node feature associated

with the spin state (σ_i) on each node. The output is a binary classification of whether the graph describes a system that is below or above T_{crit} . No edge features are considered here. This architecture is nearly identical to the one presented in the documentation[5] of the PyG package for PyTorch[2]. The code for this GCNN is also available in the supplementary.

D. Test Data

Traditional ways of training and testing a machine learning model is to get a single data set, and split it into a training, validation, and test set. While this general scheme works well for most datasets, complications enter when considering data produced by the Monte Carlo calculations described above. Namely, due to possible autocorrelations between Monte Carlo steps, conducting such a training-validation-test split may not be optimal for judging how well a model is performing. To address this, the method used here modifies the test part of this scheme. A dataset is still split into training, validation, and test datasets, with each used in the traditional way to train and monitor the machine learning model. However, after the model is trained and tested against the initial test set, a new test set is used to verify the performance measured by the previous procedure. This new test set is derived from a completely separate Monte Carlo simulation, at different temperatures. This new test set thus carries no concerns over autocorrelations with the Monte Carlo data used in the training. Additionally, this data

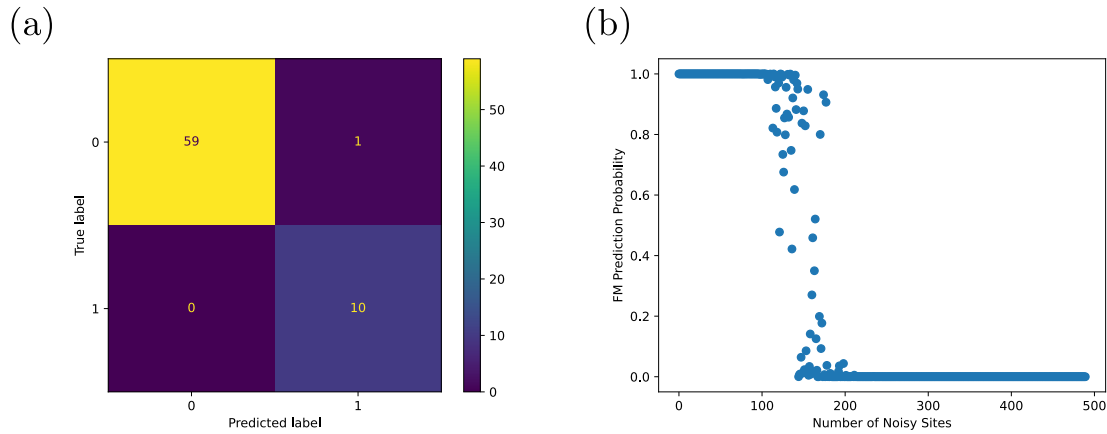


FIG. 4. Results obtained from applying a FCNN to test data (a) and artificial data (b) is shown. The FCNN was trained on PM data and FM data where only the $\sigma_i = +1 \forall i$ case is represented. While the confusion matrix (a) suggests the FCNN achieves high accuracy when predicting FM and PM states, the analysis on artificial data (b) shows the contrary. Specifically, it shows that when the system predominantly becomes -1 spins (large number of “Noisy Sites”), the model fails to classify these states as FM.

is sampled with more Monte Carlo steps in-between. Specifically, the Monte Carlo computation is allowed 2000 (or 1000 if domain wall data) steps to thermalize, then the first test sample is taken. Subsequent samples are taken in intervals of 1×10^6 steps (for non-domain wall data) or 5×10^5 steps (for domain wall data). The larger spacing between steps should help ensure that autocorrelations between samples are minimal. Unless stated otherwise, all results reported (such as all of the confusion matrices that appear below) are obtained by applying a trained model to new test data, as defined here.

III. RESULTS

A. Phase Transition

The first step of the analysis is to identify T_{crit} , so data can be labeled as $T < T_{\text{crit}}$ or $T > T_{\text{crit}}$. While rigorous methods exist for doing this (such as Binder’s cumulant), a rough estimate of the transition temperature is taken from the magnetization plot shown in Fig. 5. The critical temperature determined from Fig. 5 is $T_{\text{crit}} \approx 0.9$.

B. Spontaneously-Broken Symmetry Initial Condition

The equivalence of the $\sigma_i = +1 \forall i$ and $\sigma_i = -1 \forall i$ FM cases is an example of spontaneous symmetry breaking. While a physicist can clearly see that both cases, while visually in-equivalent, are physically identical, a machine learning model that is not privy to symmetry may have trouble analyzing these states.

In order to investigate this, the FCNN described in the previous section is trained on two different training

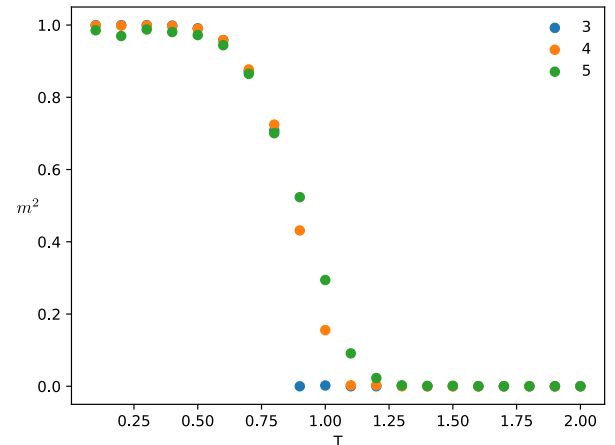


FIG. 5. Magnetization-squared as a function of temperature for a $\{10, 3\}$ hyperbolic lattice with various numbers of generations (see legend). At low temperatures (roughly $T < 0.9$) $m^2 \approx 1$, indicating a FM order. Meanwhile, at high temperatures (roughly $T > 0.9$) $m^2 \approx 0$, indicating PM order. From this plot, the critical temperature is determined to be roughly at $T_{\text{crit}} \approx 0.9$. This data was obtained using the expectation-informed initial condition: $\sigma_i = +1 \forall i$.

datasets: one where the FM ($T < T_{\text{crit}}$) data is represented by $\sigma_i = +1 \forall i$ and one where the FM training data includes both cases of the order.

Starting with the case where only one of the two equivalent FM cases is represented in the training data, the FCNN is shown to have good predictive power on test data where the FM ordered phases are also limited to $\sigma_i = +1 \forall i$. This result is shown explicitly by the confusion matrix in Fig.4. However, the breakdown of this machine learning model is easily demonstrated by feeding it artificial data, as shown in Fig. 4. Here, artificial

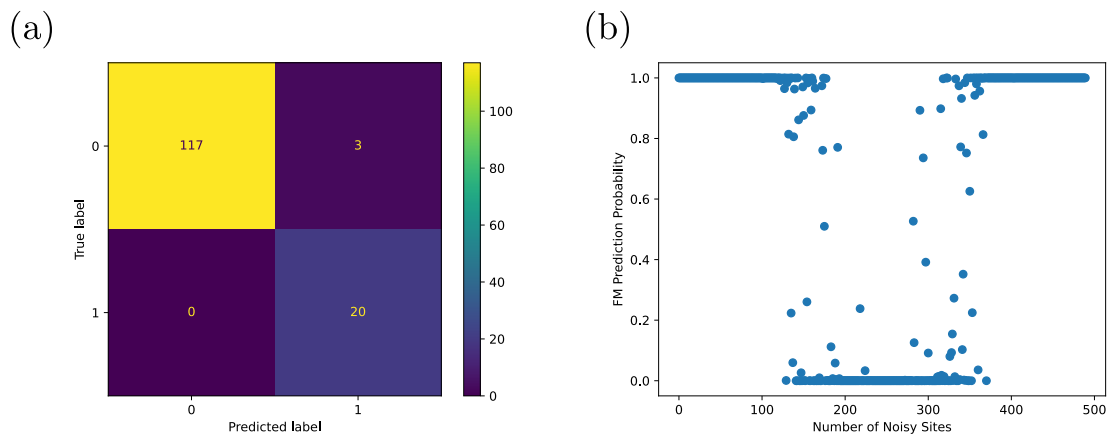


FIG. 6. Results from applying a FCNN trained on PM data and FM data ($\sigma_i = +1 \forall i$ or $\sigma_i = -1 \forall i$) to test data (a) and artificial data (b). The confusion matrix (a) shows that the model performs well against the test data. From looking at the artificial data (b), it is clear that, unlike the original FCNN trained on only a subset of possible FM data, the newly-trained FCNN can recognize the FM order when the system is predominantly composed of $+1$ or -1 spins.

data with all spins set to $+1$ is generated. Then “noise” is progressively added in the form of a random site of the spin configuration being flipped to -1 (this random sampling is such that any already flipped sites are not re-rolled). This procedure continues until the system is in a state where all spins are equal to -1 . As shown, the model correctly identifies the FM order when a vast majority of the spins are $+1$, and the PM when there is no considerable consensus of $+1$ or -1 spins in the system. However, when the “noise” becomes so extreme that the system transitions to a configuration where the system is predominantly -1 spins, the machine learning model fails to see this as a FM ordered state. This indicates that the FCNN, trained on the $\sigma_i = +1 \forall i$ FM case data, appears to be learning the FM phase through the number of $+1$ spins present in the system, which is incorrect.

In order to overcome this limitation, a FCNN with the same architecture is trained on data where both $\sigma_i = +1 \forall i$ and $\sigma_i = -1 \forall i$ FM cases are represented. The predictions of this model on test data is shown in Fig. 6. As shown, the performance of this newly-trained model is high on test data where both FM cases are represented. This is the first indication that, by feeding the model more representative training data, the FCNN’s ability to identify both cases of FM is improved. To further support this claim, the same analysis with artificial data (as described above) is used on this model. The result of this analysis is shown in Fig. 6. Unlike the FCNN model trained on only one case of the FM order, Fig. 6 clearly shows that this newly-trained FCNN now recognizes the predominantly -1 spin cases as a FM. From this discussion, it is shown that the problem of a FCNN misidentifying the FM order is fixed by ensuring that the training data is representative of all cases of the FM order.

C. Domain Walls

A final component of the Ising model considered here is the phenomenon of domain walls. While ideal pictures of the FM state have all spins equal to ± 1 , localized regions of either variety of FM order can form throughout the system, with domain walls referring to the “walls” which delineate these localized “domains” of ± 1 spin. An illustrative picture of domains walls on a $\{10, 3\}$ hyperbolic lattice is shown in Fig. 7. The key questions remains: can the simple FCNN proposed above pick out a FM state from localized features like domain walls? To test this, a FCNN is trained on data with FM order represented by predominantly ± 1 states, and by FM data with domain walls. The predictions this trained model makes on test data is shown in Fig. 8. At first glance, Fig. 8

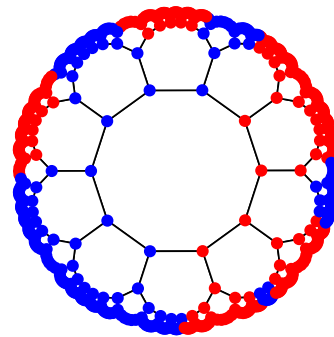


FIG. 7. Illustrative picture of magnetic domains on a $\{10, 3\}$ hyperbolic lattice. Blue and red marking represent the two Ising spins. This figure was made with the help of the python package Hypertiling[6].

seems to indicate that the FCNN retains somewhat good predictive power, even with domain walls. However, if evaluated only on test data with domains walls, the

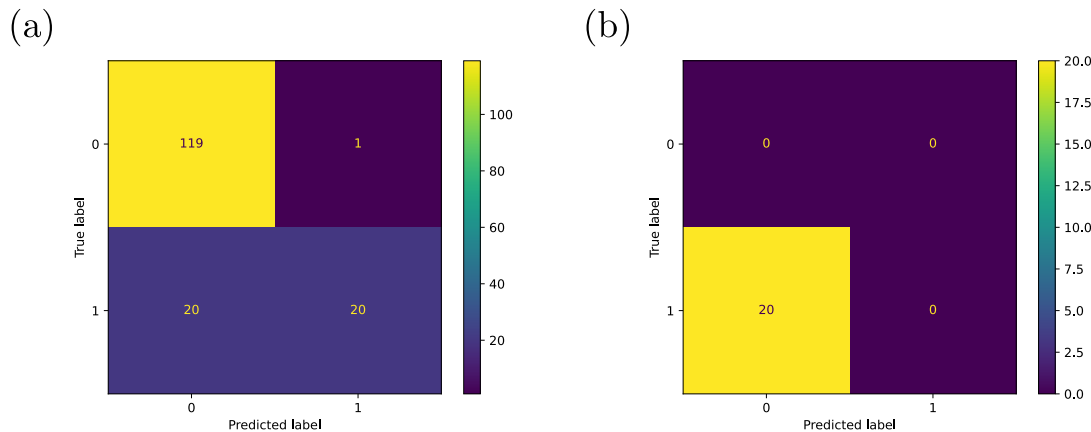


FIG. 8. Confusion matrices obtained by applying a FCNN trained on PM data and all varieties of the FM data. When used to predict test data with PM data and all FM data varieties (a), the FCNN shows a clear performance degradation compared to the results obtained in Fig. 6. To confirm that the cause is the domain wall FM data, the FCNN is used on test data only featuring the FM domain wall data (b). It is clear that the vast majority of the mistakes the model can be credited to the domain wall data.

confusion matrix shown in Fig. 8 is obtained. It is clear from Fig. 8(a) and Fig. 8(b) that the newly-trained model still predicts FM states with predominantly ± 1 spins well, however the model's predictive power is greatly reduced when considering domain walls. More specifically, the FCNN fails completely when trying to classify FM data with domain walls, despite being trained on similar data. Interestingly, when evaluated on the original test data (the test data which was split from the same Monte Carlo run as the training and validation data), the trained FCNN had a loss value of 0.014 and accuracy of 0.998, indicating that the trained model did correctly classify the domain wall data in that test set. There are two possibilities here: the test data could have been split such that no domain wall data was present, or the autocorrelations of the domain wall data are long, thus causing the model to seem to have performed much better than it actually does. The former is unlikely since equal amounts of data from all three cases of initialization were utilized to create the training, validation, and test datasets. As for the latter, this seems plausible, especially considering that there are less Monte Carlo steps separating sampled data for the domain wall dataset (see the Methodology section for details). If this is the case, the initial high-performance recorded on the test data split from same Monte Carlo simulation may be accredited to the FCNN learning the specific magnetic domains featured in the training data, and which are re-expressed in the test data due to autocorrelations. This is an example of why the new testing scheme (described above) is required, when quantifying the performance of a machine learning model trained on Monte Carlo generated data.

In order to improve on the FCNN, a GCNN is considered. The GCNN is trained on PM data and FM data with all cases ($\sigma_i = +1 \forall i$, $\sigma_i = -1 \forall i$, and domain walls) represented. The performance of the trained

GCNN on test data with all varieties of FM is shown in Fig. 9. Clearly, the GCNN demonstrates strong predictive power on the test data.

To emphasize the efficacy of the GCNN at classifying FM order with domain walls, the confusion matrix obtained by applying the GCNN model to only FM data with domain walls is shown in Fig. 9. Unlike the FCNN model, the GCNN model shows good performance on the FM domain wall data. This suggests that, unlike the FCNN, the GCNN has the power to identify local features on hyperbolic lattices.

IV. CONCLUSIONS

A. Summary

This work serves as a systematic study of the efficacy of FCNN and GCNN models applied to the Ising model on a hyperbolic lattice. It was shown that the FCNN can capture much of the physics of the Ising model when the FM state is not defined over localized, magnetic domains. Additionally, the sensitivity of the FCNN to the training data was also exposed by the juxtaposition of the FCNN performance when trained on FM data that had varying amounts of representation of the $\sigma_i = +1 \forall i$ and $\sigma_i = -1 \forall i$ cases. In order to resolve the outstanding problem of the FCNN's failure to properly classify FM data with domain walls, a GCNN model was introduced. Unsurprisingly, the GCNN showed improved performance over the FCNN, succeeding in classifying the FM ordered phase even in the presence of domain walls.

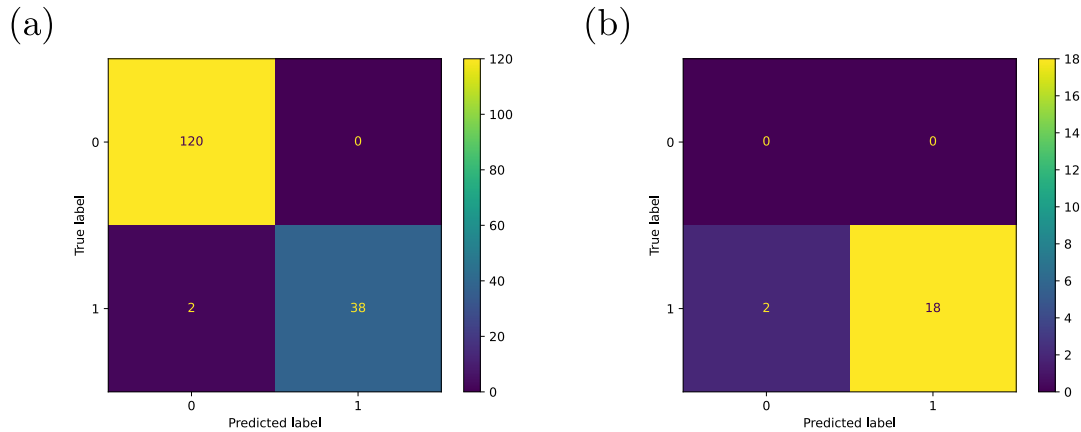


FIG. 9. Confusion matrix obtained by applying a GCNN trained on PM data, and all three varieties of FM data. Data included in the test dataset were the PM case and all FM cases (a) or just FM data with domain walls (b). These results exemplify the high-performance of the GCNN to classify all studied varieties of Ising model data on a hyperbolic lattice.

B. Future Work

While the Ising model is a surprisingly interesting model despite its simplicity, other quantum spin models will be interesting to look at. Namely, it will be interesting to look at the XY model (the next higher dimensional analog to the Ising model) on hyperbolic lattices. There exists studies in the literature that have researched into machine learning models' abilities to recognize the Kosterlitz-Thouless transition[3, 4]. However, these studies have focused on traditional machine learning models like FCNNs and convolutional neural networks, since the systems they were studying were Euclidean lattices.

It will be interesting to see how these studies can be extended to more complex systems like hyperbolic lattices, and more complex machine learning models like the GCNN studied here.

C. Acknowledgment

Other studies focused on the Ising model on hyperbolic lattices exists[7, 8]. In this way, the study of the Ising model on these systems is not novel. However, to my knowledge, this is the first study to use machine learning methods to analyze Ising model results on a hyperbolic lattice.

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